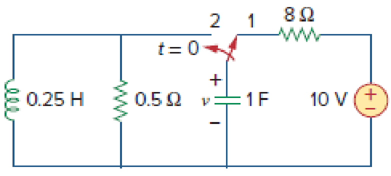
Problem

In the circuit of Fig. 16.76, the switch has been in position 1 for a long time but moved to position 2 at $t = 0$. Find:

(a) $v(0+)$, $dv(0+)/dt$

(b) $v(t)$ for $t \geq 0$.

Figure 16.76:



Step-by-step solution

Step 1 of 7

(a)

Refer to Figure 16.76 in the textbook.

At $t<0$, the switch has been in position 1 for a long time. Thus, the circuit was in steady state before $t=0$.

An inductor behaves as a short circuit and a capacitor behaves as an open circuit in steady state.

Redraw the Figure for $t<0$ as shown in Figure 1.

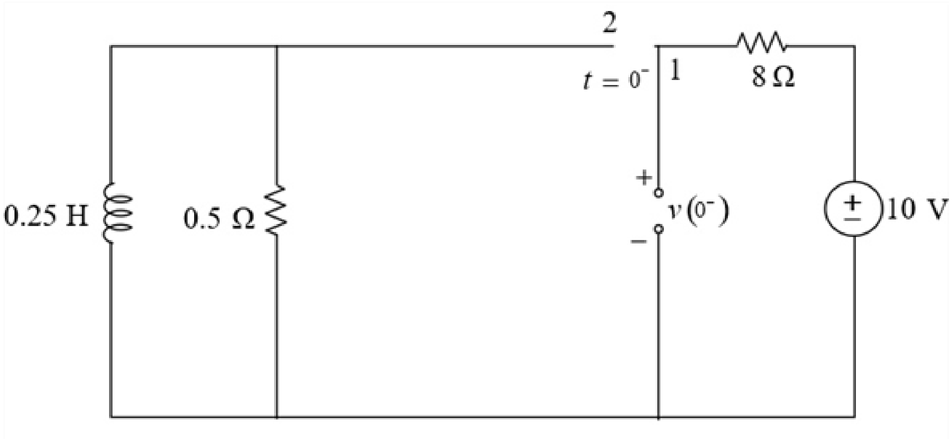


Figure 1

Step 2 of 7

The initial current through the inductor is zero.

$$i_L(0^-) = 0 \text{ A}$$

Calculate the initial voltage across the capacitor.

$$10 - 8(i) - v(0^-) = 0$$
$$10 - 8(0) - v(0^-) = 0$$
$$v(0^-) = 10 \text{ V}$$

Capacitor does not allow sudden change in the voltage, that is,

$$v(0^+) = v(0^-)$$

Substitute 10 V for $v(0^-)$.

$$v(0^+) = 10 \text{ V}$$

Hence, the value of $v(0^+)$ is $\boxed{10 \text{ V}}$.

Step 3 of 7

Redraw the circuit in Figure 1 for the switch in position 2 as shown in Figure 2.

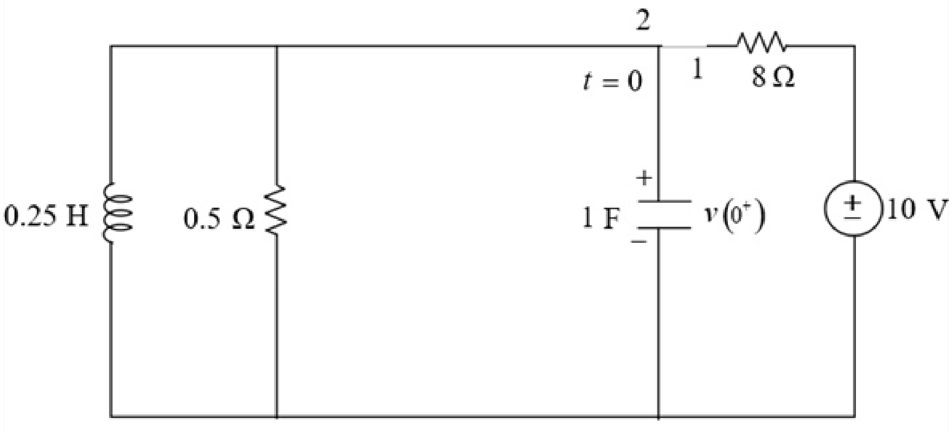


Figure 2

Step 4 of 7

Apply Kirchhoff's Current Law in Figure 2.

$$1\frac{dv(0^+)}{dt} + \frac{v(0^+)}{0.5} + i_L(0^+) = 0$$

Substitute 10 V for $v(0^+)$ and 0 A for $i_L(0^+)$.

$$1\frac{dv(0^+)}{dt} + \frac{10}{0.5} + 0 = 0$$
$$\frac{dv(0^+)}{dt} = -20 \text{ V/s}$$

Hence, the value of $\frac{dv(0^+)}{dt}$ is $\boxed{-20 \text{ V/s}}$.

Step 5 of 7

(b)

Redraw the circuit in Figure 2 in s-domain as shown in Figure 3.

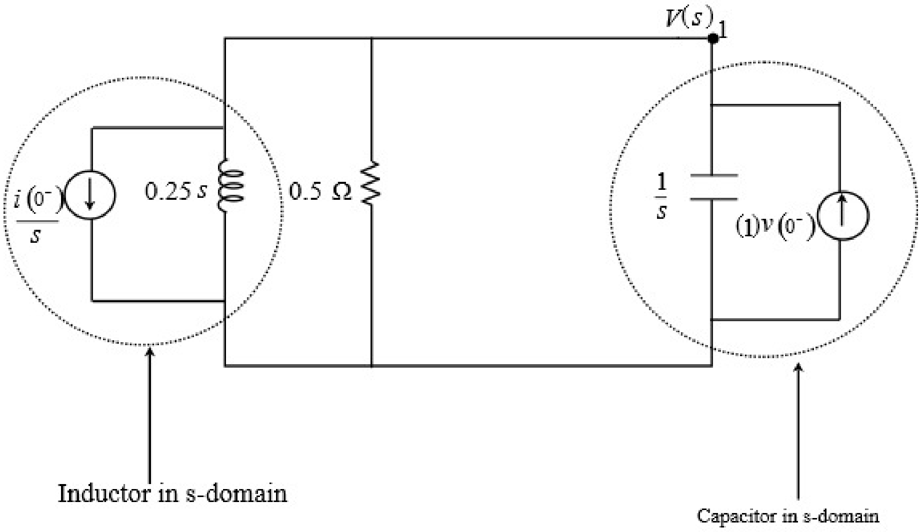


Figure 3

Step 6 of 7

Apply Kirchhoff's Current Law at node 1.

$$\frac{V(s)}{\left(\frac{1}{s}\right)} - v(0^-) + \frac{V(s)}{0.5} + \frac{V(s)}{0.25s} + \frac{i(0^-)}{s} = 0$$

Substitute $10\,\text{V}$ for $v(0^-)$ and $0\,\text{A}$ for $i(0^-)$.

$$sV(s) - 10 + 2V(s) + 4\frac{V(s)}{s} = 0$$

$$\left(s + 2 + \frac{4}{s}\right)V(s) = 10$$

$$\left(\frac{s^2+2s+4}{s}\right)V(s) = 10$$

$$\begin{aligned} V(s) &= \frac{10s}{s^2+2s+4} \\ &= \frac{10s}{(s+1)^2+\left(\sqrt{3}\right)^2} \\ &= \frac{10(s+1)-10}{(s+1)^2+\left(\sqrt{3}\right)^2} \\ &= \frac{10(s+1)}{(s+1)^2+\left(\sqrt{3}\right)^2} - \frac{5.773\sqrt{3}}{(s+1)^2+\left(\sqrt{3}\right)^2} \end{aligned}$$

Step 7 of 7

Apply inverse Laplace transform.

$$\begin{aligned} v(t) &= 10\mathrm{L}^{-1}\left[\frac{(s+1)}{(s+1)^2+\left(\sqrt{3}\right)^2}\right]-5.773\mathrm{L}^{-1}\left[\frac{\sqrt{3}}{(s+1)^2+\left(\sqrt{3}\right)^2}\right] \\ &= 10e^{-t}\cos(1.732t)u(t)-5.773e^{-t}\sin(1.732t)u(t) \\ &= e^{-t}\left[10\cos(1.732t)-5.773\sin(1.732t)\right]u(t)\text{ V} \end{aligned}$$

Consider $10 = r\cos\theta$ and $5.77 = r\sin\theta$.

$$r^2\cos\theta + r^2\sin\theta = 10^2 + 5.77^2$$

$$r^2 = 132.49$$

$$r = 11.5$$

$$\frac{r\sin\theta}{r\cos\theta} = \frac{5.77}{10}$$

$$\tan\theta = 0.577$$

$$\theta = 30°$$

Substitute $11.5\cos 30°$ for 10 and $11.5\sin 30°$ for 5.77 in the voltage expression.

$$\begin{aligned} v(t) &= e^{-t}\left[11.5\cos(30°)\cos(1.732t)-11.5\sin(30°)\sin(1.732t)\right] \\ &= 11.5e^{-t}\left[\cos(30°)\cos(1.732t)-\sin(30°)\sin(1.732t)\right] \\ &= 11.5e^{-t}\cos(1.732t+30°)\text{ V} \end{aligned}$$

Therefore, the value of voltage $v(t)$ is $\boxed{\left[11.5e^{-t}\cos(1.732t+30°)\right]\text{V}}$.